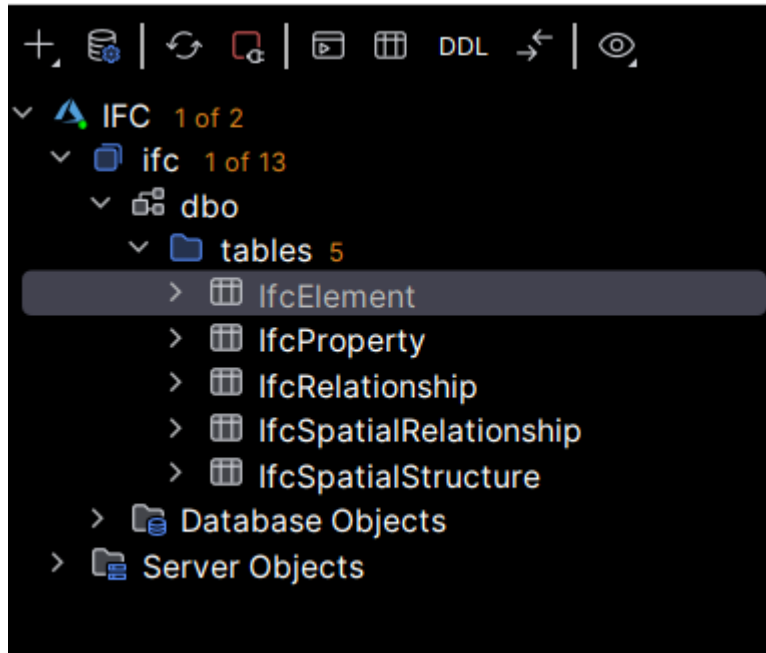


## IFC Database Architecture:

Generally there are 5 tables considered to include an entire IFC File. They are namely IfcElements, IfcProperty, IfcRelationship, IfcSpatialRelationship and IfcSpatialStructure. As the next step either the IFC Elements and Spaces tables will be merged into one making the database working with 3 whole tables, or a spaces property table will be added as well



## Further details:

### Database Schema:

Table: database\_firewall\_rules

- Column: id (int)
- Column: name (nvarchar)
- Column: start\_ip\_address (varchar)
- Column: end\_ip\_address (varchar)
- Column: create\_date (datetime)
- Column: modify\_date (datetime)

Table: IfcElement

- Column: ElementID (int)
- Column: GlobalID (varchar)
- Column: Name (varchar)
- Column: Type (varchar)

Table: IfcProperty

- Column: PropertyID (int)
- Column: ElementID (int)
- Column: PropertyName (varchar)
- Column: PropertyValue (varchar)

Table: IfcRelationship

- Column: RelationshipID (int)
- Column: SourceElementID (int)
- Column: TargetElementID (int)
- Column: RelationshipType (varchar)

Table: IfcSpatialRelationship

- Column: SpatialRelationshipID (int)
- Column: ParentSpatialID (int)
- Column: ChildSpatialID (int)
- Column: RelationshipType (varchar)

Table: IfcSpatialStructure

- Column: SpatialID (int)
- Column: GlobalID (varchar)
- Column: Name (varchar)
- Column: Type (varchar)

Primary Keys:

- IfcElement: ElementID
- IfcProperty: PropertyID
- IfcRelationship: RelationshipID
- IfcSpatialRelationship: SpatialRelationshipID
- IfcSpatialStructure: SpatialID

Foreign Keys:

- IfcProperty.ElementID -> IfcElement.ElementID
- IfcRelationship.SourceElementID -> IfcElement.ElementID

- IfcRelationship.TargetElementID -> IfcElement.ElementID
- IfcSpatialRelationship.ChildSpatialID -> IfcElement.ElementID
- IfcSpatialRelationship.ParentSpatialID -> IfcSpatialStructure.SpatialID

#### Valid Values:

- IfcElement.Type: IfcCurtainWall, IfcFurniture, IfcSanitaryTerminal, IfcStair, IfcBuildingElementPart, IfcLightFixture, IfcRailing, IfcWindow, IfcOpeningElement, IfcWall, ...
- IfcElement.Name: Aluminium, Ausbauplatte GKB, Barhocker, Barpult, Büroarbeitsplatz, Büroplanung, Bürotrennwand, Couchtisch, Doppelwaschbecken, Duschabtrennung, ...
- IfcProperty.PropertyName: netsidearea, height, einheit, isexternal, festigkeitsklasse estrich, requiredheadroom, estrichart, grosssidearea, feuerwiderstandsklasse, netarea, ...
- IfcRelationship.RelationshipType: IfcRelAggregates, IfcRelAssociatesMaterial, IfcRelContainedInSpatialStructure, IfcRelDecomposes, IfcRelDefinesByProperties, IfcRelVoidsElement
- IfcSpatialRelationship.RelationshipType: IfcRelContainedInSpatialStructure
- IfcSpatialStructure.Name: EG, Gebäude, Gelände, OG1, OG2, R-006, R-007, R-008, R-009, R-010, ...
- IfcSpatialStructure.Type: IfcBuilding, IfcBuildingStorey, IfcSite, IfcSpace

- Below is an example view of **IfcElements** table:

| WHERE |           | ORDER BY                 |             |                        |
|-------|-----------|--------------------------|-------------|------------------------|
|       | ElementID | GlobalID                 | Name        | Type                   |
| 1     |           | 05rUuJDRn9m9ZDHFfuduY3   | Wand-001    | IfcWall                |
| 2     |           | 0jQb_ogX7armJc6jZIwbIH   | Gipsputz    | IfcBuildingElementPart |
| 3     |           | 24\$DBaelw0kKuaMtKKziw7  | Stahlbeton  | IfcBuildingElementPart |
| 4     |           | 0jYsvC__s5wj3Ygs7uNpc2   | XPS         | IfcBuildingElementPart |
| 5     |           | 2\$6WkoTSkn14MDbWILVV0x  | Vollziegel  | IfcBuildingElementPart |
| 6     |           | 37d9j\$X30CrUU_2B4pt45L  | Fenster-001 | IfcOpeningElement      |
| 7     |           | 0yD_0pik7\$ZGHe5w6TlvsV4 | Fenster-001 | IfcOpeningElement      |
| 8     |           | 0p07_yV18Q0daIJQC4ST5h   | Fenster-002 | IfcOpeningElement      |
| 9     |           | 3s_zmCBwz1KwauAhYsD\$WC  | Fenster-001 | IfcWindow              |
| 10    |           | 16e0lg\$Sz14fRAwrUSoP10  | Fenster-001 | IfcWindow              |
| 11    |           | 0PTt1B30TBpPgVbEADxnAB   | Fenster-002 | IfcWindow              |
| 12    |           | 2\$o0vCjxj6rweiaCm38UWH  | Wand-002    | IfcWall                |
| 13    |           | 2CTIKd1KUffhAivxYuTHMm   | Gipsputz    | IfcBuildingElementPart |
| 14    |           | 1rBRnz83Llee4BRMV3QfG_   | Stahlbeton  | IfcBuildingElementPart |
| 15    |           | 0cd4XY\$uv_hFAIYNMMU8C4  | Gipsputz    | IfcBuildingElementPart |
| 16    |           | 3J1LCCfAF_mF0mEV\$0WhGL  | Tür-004     | IfcOpeningElement      |
| 17    |           | 2dpfLLFRD4M0uzN206v9LW   | Tür-004     | IfcDoor                |
| 18    |           | 3oPbUP4rTDx9CcqpnVvBG0   | Wand-003    | IfcWall                |
| 19    |           | 2D6DBla4gFJZneczHSHpVz   | Gipsputz    | IfcBuildingElementPart |
| 20    |           | 3o4SaRxn_bIOGMqw7v0Hcj   | Stahlbeton  | IfcBuildingElementPart |
| 21    |           | 2bkczIBZQDcLsqCkmhm0UuH  | XPS         | IfcBuildingElementPart |
| 22    |           | 379GR1fUUfTQgVzjT7wBNB   | Vollziegel  | IfcBuildingElementPart |
| 23    |           | 083k_Le3PaglwXUKVt940S   | Fenster-003 | IfcOpeningElement      |
| 24    |           | 1MpLJraR162r0Xv2NH9xCk   | Fenster-004 | IfcOpeningElement      |
| 25    |           | 3ztJPqg0BnPyLSD1KwPmcF   | Fenster-005 | IfcOpeningElement      |
| 26    |           | 3j3fY6t8D14gCYpS9zI\$qs  | Fenster-003 | IfcWindow              |
| 27    |           | 3DGSbZcYL2nhPRPEf53ITa   | Fenster-004 | IfcWindow              |
| 28    |           | 28ye_Y7wLDCvvANMyaxKhU   | Fenster-005 | IfcWindow              |
| 29    |           | 2WXXDfXkL1380weJ9sKHxU   | Wand-004    | IfcWall                |
| 30    |           | 37h4jJyjai1cSzWAdYPfL_   | Gipsputz    | IfcBuildingElementPart |
| 31    |           | 06cAxm9PzamjkRL9Kq8Qhf   | Stahlbeton  | IfcBuildingElementPart |
| 32    |           | 1V60CodTA6ktnobxKDSG\$C  | XPS         | IfcBuildingElementPart |

Allowing the element type as a column gives us the freedom of not having to add custom columns to the table the every time a new element with its unique property shows up. Same goes for **IfcSpaces** table as shown below:

| WHERE |           | ORDER BY |                         |         |                   |
|-------|-----------|----------|-------------------------|---------|-------------------|
|       | SpatialID |          | GlobalID                | Name    | Type              |
| 1     |           | 1        | 20FpTZCqJy2vhVJYtjuIce  | Gelände | IfcSite           |
| 2     |           | 2        | 00tMo7QcxqWdIGvc4sMM2A  | Gebäude | IfcBuilding       |
| 3     |           | 3        | 2Wk49hkHPC2hvwchm5NASJ  | EG      | IfcBuildingStorey |
| 4     |           | 4        | 3UwyHsU1LB6uAfwFoS4U_u  | R-020   | IfcSpace          |
| 5     |           | 5        | 0wXKNuTiz8jw173C6\$qjNF | R-006   | IfcSpace          |
| 6     |           | 6        | 16wr6GEgH0F8ZPp7SSVFBx  | R-007   | IfcSpace          |
| 7     |           | 7        | 0Y0oa7e7P7S90vbhcSBYbu  | R-008   | IfcSpace          |
| 8     |           | 8        | 2AEVfeKQj929ks9be\$t11M | R-009   | IfcSpace          |
| 9     |           | 9        | 1yqvnUxP9Bgg8DAUUrwg1X  | R-010   | IfcSpace          |
| 10    |           | 10       | 1YTEnQWwL08Qrl5ab3Q1Yn  | R-011   | IfcSpace          |
| 11    |           | 11       | 1NcDDR8Kv5J90DC74NtAqL  | R-012   | IfcSpace          |
| 12    |           | 12       | 16BV5Rj71ClvYmyLCxk9gQ  | R-013   | IfcSpace          |
| 13    |           | 13       | 0M3BT4LYT9R9HwAWNqoMZ_  | R-014   | IfcSpace          |
| 14    |           | 14       | 2Fw77eRtT6Vv6lQECP_siq  | R-015   | IfcSpace          |
| 15    |           | 15       | 0ci8HWVF14o9qE1P4AtqRa  | R-016   | IfcSpace          |
| 16    |           | 16       | 19kM0PvdH5wfAEkvXo67cF  | R-017   | IfcSpace          |
| 17    |           | 17       | 3te3FaTfzCZvgE1ZBWxu6c  | R-018   | IfcSpace          |
| 18    |           | 18       | 3mDA6TNnXBe8JMWhE\$yuuN | R-019   | IfcSpace          |
| 19    |           | 19       | 0tTixgxXn8A9HicyqmhwKG  | R-022   | IfcSpace          |
| 20    |           | 20       | 2u8QLkhx4HBrLqA_wnknJ   | 0G1     | IfcBuildingStorey |
| 21    |           | 21       | 3Ud5H4_0rD2fmFA8thfn7Z  | R-023   | IfcSpace          |
| 22    |           | 22       | 2ED0B\$SLrAmvzoIimdQswm | R-024   | IfcSpace          |
| 23    |           | 23       | 3Gg7GryBnFIRY5RBRhEqSn  | R-025   | IfcSpace          |
| 24    |           | 24       | 3Br_vZW_XCChCxqFJqfADEk | R-026   | IfcSpace          |
| 25    |           | 25       | 10ItwPwRH00vKBJF1S1FJi  | R-027   | IfcSpace          |
| 26    |           | 26       | 3Az60UAA94vfvi\$8HAP1UB | R-024   | IfcSpace          |
| 27    |           | 27       | 3GLLeok3jDv8X9xXoIr6fe  | R-025   | IfcSpace          |
| 28    |           | 28       | 20M0b2liX918GE_qQKa7N   | 0G2     | IfcBuildingStorey |

IFC Properties table follows the same design as explained above:

|    | PropertyID ▾ | ElementID ▾ | PropertyName ▾         | PropertyValue ▾ |
|----|--------------|-------------|------------------------|-----------------|
| 1  | 1            | 1           | name                   | wand-001        |
| 2  | 2            | 1           | material               | stahlbeton      |
| 3  | 3            | 1           | betongüte              | c30/37          |
| 4  | 4            | 1           | expositionsklasse      | xf3             |
| 5  | 5            | 1           | überwachungsklasse     | ük1             |
| 6  | 6            | 1           | sichtbetonklasse       | sb2             |
| 7  | 7            | 1           | brandwand              | false           |
| 8  | 8            | 1           | feuerwiderstandsklasse | f30             |
| 9  | 9            | 1           | baustoffklasse         | b2              |
| 10 | 10           | 1           | id                     | 1128            |
| 11 | 11           | 1           | isexternal             | true            |
| 12 | 12           | 1           | status                 | ['EXISTING']    |
| 13 | 13           | 1           | loadbearing            | true            |
| 14 | 14           | 1           | length                 | 14.705          |
| 15 | 15           | 1           | height                 | 4.0             |
| 16 | 16           | 1           | width                  | 0.41            |
| 17 | 17           | 1           | grossfootprintarea     | 6.09229936476   |
| 18 | 18           | 1           | netfootprintarea       | 6.09229936476   |
| 19 | 19           | 1           | grosssidearea          | 58.86           |
| 20 | 20           | 1           | netsidearea            | 35.76           |
| 21 | 21           | 1           | grossvolume            | 24.3692         |
| 22 | 22           | 1           | netvolume              | 14.8982         |
| 23 | 23           | 2           | name                   | gipsputz        |
| 24 | 24           | 3           | name                   | stahlbeton      |
| 25 | 25           | 4           | name                   | xps             |
| 26 | 26           | 5           | name                   | vollziegel      |
| 27 | 27           | 6           | name                   | fenster-001     |
| 28 | 28           | 6           | depth                  | 0.41            |
| 29 | 29           | 6           | height                 | 2.2             |
| 30 | 30           | 6           | width                  | 3.5             |
| 31 | 31           | 6           | area                   | 7.7             |
| 32 | 32           | 6           | volume                 | 0.2332365       |

IfcRelationships also has the relationship type as one column as before to allow for flexibility in case new relations show up. Now the Source and Target ID style also allows for countless nested relationships to be presented here with absolute flexibility

|    | RelationshipID ▾ | SourceElementID ▾ | TargetElementID ▾ | RelationshipType ▾ |
|----|------------------|-------------------|-------------------|--------------------|
| 1  | 1                | 1                 | 2                 | IfcRelDecomposes   |
| 2  | 2                | 1                 | 3                 | IfcRelDecomposes   |
| 3  | 3                | 1                 | 4                 | IfcRelDecomposes   |
| 4  | 4                | 1                 | 5                 | IfcRelDecomposes   |
| 5  | 5                | 12                | 13                | IfcRelDecomposes   |
| 6  | 6                | 12                | 14                | IfcRelDecomposes   |
| 7  | 7                | 12                | 15                | IfcRelDecomposes   |
| 8  | 8                | 18                | 19                | IfcRelDecomposes   |
| 9  | 9                | 18                | 20                | IfcRelDecomposes   |
| 10 | 10               | 18                | 21                | IfcRelDecomposes   |
| 11 | 11               | 18                | 22                | IfcRelDecomposes   |
| 12 | 12               | 29                | 30                | IfcRelDecomposes   |
| 13 | 13               | 29                | 31                | IfcRelDecomposes   |
| 14 | 14               | 29                | 32                | IfcRelDecomposes   |
| 15 | 15               | 29                | 33                | IfcRelDecomposes   |
| 16 | 16               | 40                | 41                | IfcRelDecomposes   |
| 17 | 17               | 40                | 42                | IfcRelDecomposes   |
| 18 | 18               | 40                | 43                | IfcRelDecomposes   |
| 19 | 19               | 40                | 44                | IfcRelDecomposes   |
| 20 | 20               | 51                | 52                | IfcRelDecomposes   |
| 21 | 21               | 51                | 53                | IfcRelDecomposes   |
| 22 | 22               | 51                | 54                | IfcRelDecomposes   |
| 23 | 23               | 51                | 55                | IfcRelDecomposes   |
| 24 | 24               | 62                | 63                | IfcRelDecomposes   |
| 25 | 25               | 62                | 64                | IfcRelDecomposes   |

Local Filter For 'RelationshipType'

| <input type="checkbox"/> Value                             | Count |
|------------------------------------------------------------|-------|
| <input type="checkbox"/> IfcRelAggregates                  | 12    |
| <input type="checkbox"/> IfcRelAssociatesMaterial          | 12    |
| <input type="checkbox"/> IfcRelContainedInSpatialStructure | 3     |
| <input type="checkbox"/> IfcRelDecomposes                  | 456   |
| <input type="checkbox"/> IfcRelDefinesByProperties         | 12    |
| <input type="checkbox"/> IfcRelVoidsElement                | 5     |

Select a checkbox to filter rows

The same goes for **Spatial Relationships** table as below:

|    | SpatialRelationshipID | ParentSpatialID | ChildSpatialID | RelationshipType                  |
|----|-----------------------|-----------------|----------------|-----------------------------------|
| 1  | 1                     | 3               | 1              | IfcRelContainedInSpatialStructure |
| 2  | 2                     | 3               | 9              | IfcRelContainedInSpatialStructure |
| 3  | 3                     | 3               | 10             | IfcRelContainedInSpatialStructure |
| 4  | 4                     | 3               | 11             | IfcRelContainedInSpatialStructure |
| 5  | 5                     | 3               | 12             | IfcRelContainedInSpatialStructure |
| 6  | 6                     | 3               | 17             | IfcRelContainedInSpatialStructure |
| 7  | 7                     | 3               | 18             | IfcRelContainedInSpatialStructure |
| 8  | 8                     | 3               | 26             | IfcRelContainedInSpatialStructure |
| 9  | 9                     | 3               | 27             | IfcRelContainedInSpatialStructure |
| 10 | 10                    | 3               | 28             | IfcRelContainedInSpatialStructure |
| 11 | 11                    | 3               | 29             | IfcRelContainedInSpatialStructure |
| 12 | 12                    | 3               | 37             | IfcRelContainedInSpatialStructure |
| 13 | 13                    | 3               | 38             | IfcRelContainedInSpatialStructure |
| 14 | 14                    | 3               | 39             | IfcRelContainedInSpatialStructure |
| 15 | 15                    | 3               | 40             | IfcRelContainedInSpatialStructure |
| 16 | 16                    | 3               | 48             | IfcRelContainedInSpatialStructure |
| 17 | 17                    | 3               | 49             | IfcRelContainedInSpatialStructure |
| 18 | 18                    | 3               | 50             | IfcRelContainedInSpatialStructure |
| 19 | 19                    | 3               | 51             | IfcRelContainedInSpatialStructure |
| 20 | 20                    | 3               | 59             | IfcRelContainedInSpatialStructure |
| 21 | 21                    | 3               | 60             | IfcRelContainedInSpatialStructure |
| 22 | 22                    | 3               | 61             | IfcRelContainedInSpatialStructure |
| 23 | 23                    | 3               | 62             | IfcRelContainedInSpatialStructure |
| 24 | 24                    | 3               | 70             | IfcRelContainedInSpatialStructure |
| 25 | 25                    | 3               | 71             | IfcRelContainedInSpatialStructure |